

West Visayas State University  
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY  
La Paz, Iloilo City, Philippines

KKK: A HISTORICALLY BASED CLICK-AND-DRAG GAME

ABOUT THE PHILIPPINE REVOLUTION

An Undergraduate Thesis

Presented to the Faculty of the

College of Information and Communications Technology

West Visayas State University

La Paz, Iloilo City

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Entertainment and Multimedia Computing

by

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June 2026

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Approval Sheet

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### Acknowledgment

The researchers wish to express their deepest gratitude and appreciation to the following individuals and institutions who have contributed to the completion of this thesis:

West Visayas State University - College of Information and Communications Technology, for providing the academic environment, resources, and guidance necessary to pursue this study.

To Sir Mark Joseph J. Solidarios, their research Advisor, for their invaluable guidance, constructive feedback, and unwavering support throughout the development of this project.

The panel members, Dr. Evan C. Sumido, Ms. Janine P. Defante, Ms. Christy P. Villano, Dr. Ma. Beth S. Concepcion, and, finally, Mr. Mark Joseph J. Solidarios, for their comments and recommendations that helped refine and strengthen the manuscript and the game's overall development.

To the Faculty members, specifically Mr. Paul Noel Dequito, for sharing their expertise in game development which greatly informed the technical aspects of this study.

Their classmates and peers, who participated in the alpha and beta testing phases, offered honest feedback and

suggestions and kept the researchers motivated throughout the whole ordeal.

Their families, for their encouragement and moral support, helped sustain them during the challenges of research and development.

This thesis would not have been possible without the collective support, guidance, and inspiration of all those mentioned.

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K.L. Acuesta; K.F. Concepcion; P.E. Dureza; T.D. Espinosa, KKK: A Historically based Click-and-Drag game about the Philippine Revolution, Unpublished Undergraduate Thesis, Bachelor of Science in Entertainment and Multimedia Computing, College of Information and Communications Technology, West Visayas State University, Iloilo City, Philippines, 2026.

**Abstract**

This study presents the design, development, and evaluation of KKK: A Historically Based Click-and-Drag Game about the Philippine Revolution, a 2D educational game created to enhance historical literacy and engagement among the Filipino youth. The project responds to the declining emphasis on Philippine history in the K-12 curriculum by offering an interactive platform that combines gamification with historical accuracy. The game was developed using the Godot Engine and GDScript, employing click-and-drag mechanics to let players interact with cultural symbols, characters, and significant battles, such as San Juan del Monte, Imus, and Zapote Bridge. A developmental research method guided the process, encompassing planning, design, implementation, and evaluation phases. Usability and technical performance were assessed through alpha and beta testing, using researcher-developed Likert-scale questionnaires and open-ended feedback forms. Findings revealed that the game's core mechanics were intuitive and engaging, with sound effects

and user interface elements effectively supporting gameplay. However, issues such as inconsistent art styles, repetitive level design, and minor bugs were identified, highlighting areas for refinement. Despite these limitations, respondents recognized the game's potential as a supplementary educational tool, noting its ability to motivate learners and encourage participation in historical study. The study concludes that gamified digital media can serve as a viable platform for promoting historical awareness and civic engagement. Future iterations should focus on improving technical polish, diversifying gameplay, and strengthening narrative integration to maximize both educational impact and player experience.

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## CHAPTER 1 INTRODUCTION TO THE STUDY

### Background of the Study and Conceptual Framework

History is slowly forgotten by the younger generation. Removing Philippine History, especially the Philippine Revolution against the Spaniards, from the high school curriculum has resulted in a generation with a limited understanding of key events and figures. This gap makes the youth more vulnerable to historical revisionism, weakens their ability to critique current political narratives, and erodes civic engagement [10].

Retention of historical knowledge in the Philippines has faced challenges, especially since the removal of Philippine History as a dedicated subject in high school under the K-12 curriculum. According to education experts, this has led to weakened historical awareness among youth, increased vulnerability to disinformation, and a decline in civic consciousness and national identity formation. Students often memorize facts for exams but struggle to retain a deeper understanding or critical perspectives. This is partly due to traditional methods that emphasize remote learning over engagement [11].

Gamification that is applied in classroom teaching and learning is not new. The application of gamification set in an immersive theater is a fairly recent exploration.

Gamification and immersive theater share a common purpose, especially for 21<sup>st</sup>-century learners. This matter is of great importance in teaching subjects that are valuable but that students do not find interesting, such as history[9].

In recent years, Gamification has emerged as a powerful tool in education, leveraging the interactive and immersive nature of games to enhance learning experiences. By integrating game mechanics with educational content, learners are motivated to participate and retain information more effectively. This research explores the potential of Gamification in historical and cultural education by designing an interactive learning experience centered on Philippine history and culture.

This research aims to create a game that combines historical accuracy with engaging gameplay. It utilized a click-and-drag mechanic, allowing players to interact with cultural symbols and significant events, specifically the Philippine Revolution against the Spaniards. Through this in an interactive format, players learn about key historical moments and develop a deeper appreciation of the history in which they occurred.

### Objectives of the Study

Generally, the main objective of this study is to develop a two-dimensional, click-and-drag game centered on Philippine history, featuring well-defined game mechanics and playable levels that depict significant historical events from the period.

Specifically, it aims to:

1. Design a gameplay mechanic inspired by the Philippine Revolution of 1896, particularly focusing on a "morale" system.
2. Create and integrate 2D pixel art assets that reflect the characters and environments of the Spanish colonial era.
3. Develop three playable game levels that showcase the three main battles of the Philippine revolution: the battle of San Juan del Monte, Imus, and the Zapote bridge, with mechanics and assets using the Godot Game Engine.
4. To evaluate the technical aspects of the game prototype on the Philippine Revolution against Spanish colonial rule, including functionality, stability, usability, and player engagement through alpha and beta testing with selected volunteer testers, using researcher-developed survey forms and assessment tools. Once

technical performance is validated, the study examines secondary outcomes, including content delivery, player learning, and educational effectiveness.

### Significance of the Study

This study is significant in its attempt to combine historical education with interactive digital entertainment, specifically targeting the younger generation of Filipinos. In an era dominated by technology and digital media, traditional modes of teaching history often fail to sustain student interest and engagement. By developing a click-and-drag game centered on the Philippine Revolution, this study offers an alternative learning platform that is both informative and immersive.

Educators. May use the game as a supplementary instructional tool to enhance student participation and retention of historical facts. In contrast, students gain a more engaging and memorable way to understand complex historical events and figures.

Students and Learners. The game provides an alternative and engaging learning platform that promotes active participation in understanding historical events. Through interactive gameplay, learners may develop a deeper

appreciation of Philippine history while improving comprehension and retention.

Game Developers. The study offers insights into the design and implementation of educational games, particularly in integrating gamification mechanics with historical content. It may serve as a reference for future developers aiming to create similar games.

Researchers. This study contributes to the existing body of knowledge on gamification and educational technology, particularly in the context of history education. It may serve as a basis for further studies exploring the effectiveness of game-based learning and interactive systems.

Cultural Institutions and Historians. The findings highlight the potential of digital games as tools for promoting historical awareness and cultural preservation. The study demonstrates how interactive media can be utilized to make historical content more accessible and engaging to a wider audience.

### Delimitations of the Study

This study is delimited to the design, development, and initial evaluation of a 2D click-and-drag educational game prototype. The focus is on the technical aspects of game development, specifically:

Game Engine & Tools. The project utilizes Godot for development, with supporting software such as FL Studio for Audio Editing and Aseprite for Pixelation.

Game Mechanics. Limited to click-and-drag interactions for object manipulation, navigation, and progression. This constraint ensures simplicity in design and usability testing.

Graphics & Assets. Restricted to 2D sprite-based visuals, optimized for performance and accessibility.

System Performance. Evaluation includes stability (bug and crash testing) and responsiveness.

Usability Testing. Alpha and beta testing assess interface clarity, control responsiveness, and overall player experience.

Content Integration. While the game incorporates historical events (San Juan del Monte, Imus, Zapote Bridge), narrative and educational content are secondary to the validation of technical functionality.

Additionally, the testing phase involved a small group of 5 to 10 participants, primarily selected from the intended target demographic. The feedback gathered was used to evaluate the overall user experience and identify any overlooked bugs or technical issues.

#### Definition of Terms

Alpha Testing. Standard phases in software/game development are used to identify issues and gather feedback[14].

In this study, this refers to Internal testing to catch functional issues in the Philippine Revolution game.

Beta Testing. A phase of software/game testing where a product is released to a limited group of external users to evaluate usability, stability, and overall performance before final release[14].

In this study, this refers to the phase in which selected participants play the game, provide feedback on usability and engagement, and suggest improvements to historical accuracy and gameplay mechanics.

Bug. Any error or flaw in a system's code or design[14].

In this study, this refers to any gameplay malfunction, such as drag errors, visual glitches, or broken logic.

Click-and-Drag Mechanic. A common method of user interaction in digital interfaces and games[15].

In this study, this refers to the main way players interact with revolutionary figures, objects, or events in your game.

Engagement. The psychological involvement and motivation of a player[16].

In this study, this refers to measured player feedback, time spent playing, and recall of historical facts after gameplay.

Gamification. The use of game elements (points, rewards, challenges) in non-game contexts to enhance learning or motivation[17].

In this study, this refers to integrating objectives, rewards, and progression to motivate players to learn about the Philippine Revolution.

GDScript. A scripting language for the Godot Engine, similar to Python[18].

In this study, this refers to the language used to

implement all game logic, including character behavior and drag interactions.

Godot Engine/Node. Tools and structures for building interactive applications[19].

In this study, this refers to the platform and building blocks for designing and running your 2D educational game.

Scene. A collection of nodes organized together to represent a unit of functionality, such as a level, character, or interface. It can be reused, instanced, and combined with other scenes to build complex systems[20].

In this study, this refers to the structured set of nodes that make up the interactive parts of the game, such as the battlefield layout, character interactions, or educational tasks in which players drag historical figures and objects.

Script. A set of instructions written in a programming or scripting language that automates tasks, controls behaviors, or defines interactions within a system[21].

In this study, this refers to the written code (primarily in GDScript) that defines how characters behave, how objects respond to dragging, and how the game's logic ensures historical events are represented

interactively and accurately.

UI/Usability. Principles of design that determine how users interact with and navigate systems[22].

In this study, this refers to buttons, panels, and drag mechanics that determine how easily players can navigate and interact with the game.

User Experience. The overall emotional and practical response of users when interacting with a system[23].

In this study, this refers to players' overall satisfaction, efficiency, and emotional responses while learning history through gameplay.

Prototype. A preliminary version of a product used for testing and refinement[24].

In this study, this refers to the early version of the game used to test mechanics and educational effectiveness.

## CHAPTER 2 REVIEW OF RELATED STUDIES

### Review of Existing and Related Studies

#### *Current Systems*

User Acceptance Testing (UAT) and Usability Testing are two methods commonly used in evaluating software or systems. User Acceptance Testing (UAT) focuses on overall system acceptance, whereas Usability Testing specifically evaluates the user's experience when interacting with a product. Both testing approaches are essential to ensuring software quality and user satisfaction[6]. These methods were applied to evaluate the *Little Panda's Forest Animals* game, which involved 106 respondents from two campuses. The study aimed to compare the results of Usability Testing and User Acceptance Testing, following research stages that included a literature review, questionnaire development, data collection, data analysis, and conclusion formulation. Findings revealed a difference in outcomes, with Usability Testing yielding lower scores compared to User Acceptance Testing. Specifically, Usability Testing produced scores ranging from 65 to 84, interpreted as acceptable, while User Acceptance Testing results ranged from 81% to 100%, corresponding to a very good rating[6].

### *Related Systems or Solutions*

The use of game engines and development tools plays a crucial role in creating interactive educational systems. Godot Engine has been adopted in classrooms worldwide due to its open-source nature, zero licensing costs, and browser-based accessibility, making it a practical choice for student-led projects in educational media and gamification[12]. Its emphasis on usability and beginner-friendly resources, such as GDQuest's Learn GDScript From Zero, further supports its integration into curricula focused on interactive learning. Godot Engine, as an open-source platform, offers flexibility and ease of use for developing both entertaining and educational games.

A study was conducted on the open-source game development tool, Godot Engine, in the context of the growing popularity of independent game development and the increasing availability of development software[13]. The study explored the potential of Godot Engine by analyzing its design, features, and workflows. Evaluation was carried out through multiple approaches, including a comparative analysis with widely used game engines such as Unity and Unreal Engine, as well as an examination of its history, community support, and core functionalities. The

study highlighted Godot Engine's capabilities in graphics, physics, programming, animation, audio, and networking. In addition, a practical project was developed using the engine to provide a detailed understanding of its project structure and common development workflows. The outcomes of the project were thoroughly analyzed, with specific examples related to design, programming, and feature implementation. The completed project was also released as open-source software to serve as a reference for further study.

The study concluded with a discussion of the strengths of the Godot Engine as a tool for independent game development, along with reflections on the practical project and considerations for future developments and emerging trends in the field[13].

#### *Related Studies*

The increasing integration of digital technologies in education has emphasized the importance of digital literacy. Educational games are effective tools for enhancing digital literacy skills, including critical thinking, collaboration, and information processing. Their systematic review found that game-based learning has been widely implemented in various countries to support these

competencies[1]. This study examines the development of educational games to enhance digital literacy through a systematic review of the literature. Using data from the Scopus database, 189 articles were identified with the keywords "educational games in digital literacy learning," of which 13 met the inclusion criteria. The findings show that educational games have been widely implemented across countries, using various game types to improve digital literacy skills, including collaboration, creativity, information retrieval, critical thinking, and social understanding[1].

Gamification is the use of game design elements and mechanics outside video games to engage and motivate people to achieve a set goal. It has been shown to be an effective teaching strategy in various educational settings, including primary and secondary schools, colleges and universities, and corporate training programs. It was emphasized that within the Philippine educational system, gamification can serve as an effective strategy to enhance the learning experience and promote active student participation. One approach involves integrating game-based learning activities into the curriculum, which can be designed to teach specific subjects such as mathematics, science, and language arts

in an engaging, interactive manner[2]. For instance, learners may progress through levels and earn rewards by solving subject-related tasks, thereby reinforcing their skills while maintaining motivation. In addition, gamification may be implemented through educational games and applications, either as a supplement to traditional instruction or as independent learning tools. These resources can address a wide range of subjects and be adapted to various age groups and learning preferences. Furthermore, gamification can support the development of collaboration and teamwork skills through structured group activities, challenges, and competitions that encourage cooperative learning and problem-solving. However, the implementation of gamification in the Philippine context may be constrained by limited access to technology and educational resources, as many schools lack sufficient devices such as computers or tablets. Despite these limitations, educators can still adopt modified game-based strategies that are suitable for low-resource environments. Gamification is identified as a promising instructional approach in the Philippine educational system, as it not only increases student engagement and motivation but also fosters essential skills such as collaboration and active participation in the learning

process[2].

The integration of gamification into educational settings has gained recognition for its potential to boost student motivation, engagement, interest, and learning outcomes. Despite its popularity, research on gamification has produced mixed results regarding student learning outcomes. A meta-analysis was conducted to synthesize existing empirical evidence on the effectiveness of gamification as a tool for teaching and learning in educational environments[3]. Forty-one studies with 49 independent samples involving more than 5,071 participants were included in their analysis. Results from random-effects models showed a significant, large overall effect size. The researchers conducted a moderator analysis to examine how several factors affect the relationship between gamification and student learning outcomes. The study uncovered significant moderating effects for user type, educational discipline, and design principles for educational gamification, duration of "gameful" experience, and learning environment. However, measurement of student outcomes and publication type did not appear to have any significant moderating effect. Those findings have important implications for improving and implementing gamification to promote teaching and learning in future

research[3].

Gamification is an educational strategy that makes learning fun and keeps students engaged, helping them learn more and expand their knowledge. However, given its growing popularity, it is necessary to review empirical studies on gamification in education as an intervention to support effective teaching and learning. A meta-analysis to examine the impact of gamification on students' learning achievement across five moderator variables[4]. Fifteen studies published between 2018 and 2022 were included in the review based on the inclusion and exclusion criteria. The MetaEssentials v. 1.5 was used to determine effect sizes and to generate the forest plot, funnel plot, Begg-Mazumdar test, and moderator analysis. Findings revealed that gamification had a significant positive effect on student learning achievement. The five moderator variables were found to be statistically insignificant. The results suggest that gamification would be appropriate across academic courses at all educational levels in Asian regions. Moreover, Kahoot, a gamified platform with the largest effect size, could be used in well-designed in-person, online, or blended learning classes[4].

The gamification of education can enhance students'

engagement, similar to games, to improve their specific skills and optimize their learning. On the other hand, the effectiveness of gamification may vary depending on user characteristics. It was stated that the relationship among the user's characteristics, executed actions, and game elements remains an open question. To gain insights into this issue, the researchers investigated the effects of gamification on students' learning, behavior, and engagement, as influenced by their personality traits, in a web-based programming learning environment[5]. They conducted an experiment for 4 months with 40 first-year undergraduate students in programming courses. Students were randomly assigned to one of two versions of the programming learning environment: a gamified version featuring ranking, points, and badges, and the original non-gamified version. The researchers found evidence that gamification affected users in distinct ways based on their personality traits. Our results indicate that the effect of gamification depends on users' specific characteristics[5]. In the context of serious games, usability evaluation is essential. It was highlighted that serious games are increasingly recognized as effective tools in educational contexts for providing engaging,

interactive learning experiences. However, existing usability evaluation instruments often fail to adequately address the unique features and pedagogical objectives of serious games, thereby limiting their effectiveness in supporting iterative design improvements[5]. To address this gap, the study introduced the Serious Games Usability (SGU) questionnaire, developed through a systematic literature review following the methodology of Kitchenham and Charters.

The proposed instrument is a standardized tool designed to assess key usability dimensions, including gameplay mechanics, interface design, narrative engagement, and gamification elements. It was applied across multiple serious game case studies in educational settings, enabling the identification of common usability issues and context-specific design challenges. The findings demonstrated the questionnaire's relevance, clarity, and diagnostic capability, highlighting its practical value in evaluating and improving serious games. Overall, the study contributes to the field by providing a structured and context-sensitive approach to usability assessment, thereby enhancing the user experience in educational game design[7].

Game-based learning has also been explored in history education. The study conducted a comprehensive literature

review to examine the potential and challenges associated with integrating game-based learning (GBL) into formal history education. In response to the growing interest in the educational value of games, the study systematically analyzed academic publications from the past fifteen years. The review focused on two primary themes: (1) the development and theoretical foundations of history-related game-based learning, and (2) the challenges encountered in its implementation within formal education systems including issues related to curriculum alignment, teacher preparedness, and assessment practices[8]. Based on an analysis of 118 high-impact studies, the review identified both the pedagogical advantages and structural limitations of incorporating historical games into classroom instruction. The findings suggest that while GBL has the potential to enhance student engagement, motivation, and understanding of historical concepts, its effective implementation requires careful instructional design, adequate resources, and alignment with educational standards. The study concludes by offering strategic recommendations to support the future integration of game-based learning in history education. As a literature review, the research relies on synthesizing existing studies rather than collecting primary empirical data[8].

## CHAPTER 3 RESEARCH DESIGN AND METHODOLOGY

### Description of the Proposed Study

This study involved the conceptualization, design, and development of a 2D click-and-drag game. It aimed to serve as an innovative educational tool to promote historical literacy and engagement among Filipino youth. By integrating interactive gameplay with historically accurate content, the study seeks to present significant events of the Philippine Revolution in a way that is both informative and compelling.

### Methods and Proposed Enhancements

#### *Development Process*

To achieve the objectives of this study, a Developmental Research Method was employed, specifically the Design and Development Model. This approach was appropriate for projects involving the creation of digital tools and interactive systems such as educational games. The development process followed a structured pipeline that included planning, design, implementation, and evaluation stages.

Planning and Research. This initial phase focused on gathering, verifying, and synthesizing historical data to

serve as the foundation for the game's narrative and world-building. The development team ensured that the game's depiction of the Philippine Revolution is both accurate and respectful of its historical significance.

Key battles such as San Juan del Monte, Imus, and Zapote Bridge are identified as major narrative milestones. These events are contextualized not only through their tactical and political importance but also through the lived experiences of revolutionaries, civilians, and local communities. Each battle was adapted into gameplay sequences that reflect strategic challenges, emotional stakes, and evolving player objectives.

Beyond military engagements, this phase also explores notable figures such as Andres Bonifacio, Emilio Aguinaldo, and Gregoria De Jesus, whose stories inform character arcs and dialogue. Symbolic elements such as flags, uniforms, and revolutionary slogans are included and influence visual design and in-game artifacts.

The goal is to create a historically grounded yet emotionally resonant experience in which players not only engage with pivotal moments but also gain insight into the broader struggle for independence. This research directly informs level design, character development, environmental

storytelling, and mission structure.

Game Design and Asset Creation. This phase focuses on crafting the game's identity, aligning aesthetic choices with both historical immersion and player accessibility. The game utilizes 2D animations and pixel art techniques to evoke a stylized yet evocative representation of the Philippine Revolution era. This visual approach balances nostalgic charm with expressiveness. detail, enabling dynamic storytelling within a compact, scalable format.

To complement the visuals, a carefully curated soundscape was developed, including Ambient background music, sound effects that reinforce player actions, environmental cues, and emotional beats to enhance narrative delivery and engagement.

The user interface was designed with clarity and ease of use in mind, featuring intuitive layouts, readable typography, and culturally resonant iconography.

Interactive mechanics are centered on click-and-drag functionality, enabling players to manipulate game elements, such as troop movements, in a tactile and immersive way. These mechanics are chosen to foster a sense of agency and connection to the historical narrative, encouraging strategic thinking.

Overall, the design philosophy emphasizes historical

engagement through visual storytelling and culturally grounded aesthetics, gameplay simplicity to accommodate a wide range of players, including younger audiences and those new to strategy or narrative-driven games, and modular asset creation that allows for future expansions, community contributions, or educational adaptations.

Game Development. The game was developed using the GODOT Game Engine, selected for its open-source flexibility, efficient 2D rendering capabilities, and active developer community. Core gameplay systems, user interface logic, and interactive mechanics are implemented using GDScript, Godot's high-level, dynamically typed programming language designed for rapid development and tight integration with the engine's scene system. GDScript was used to script player interactions, including click-and-drag mechanics, dialogue triggers, and mission progression, manage game states and transitions between levels, ensuring smooth narrative flow aligned with historical chronology, handle UI responsiveness and accessibility features, such as tooltips, contextual prompts, and adaptive layouts, and implement educational logic, such as tracking learning objectives and triggering feedback based on player choices.

Game levels are structured to reflect the sequence of

historical events from the Philippine Revolution. Each level was designed as a self-contained scene or node hierarchy, allowing modular development and easy iteration. These levels integrate learning objectives aligned with specific historical content, ensuring that players engage with key concepts through gameplay rather than passive exposition.

For example, a level based on the Battle of Zapote Bridge might include strategic troop-placement mechanics scripted in GDScript to simulate tactics, environmental storytelling elements that reveal cultural and political context through exploration, and conditional events that unlock historical insights based on player decisions or performance.

This approach ensures that the game functions not only as an engaging interactive experience but also as a meaningful educational tool. The use of GDScript allows for agile development, precise control over gameplay behavior, and seamless integration of narrative and pedagogical elements.

Testing and Feedback. Two phases of testing were conducted: Alpha Testing (internal evaluation) and Beta Testing (external user feedback). A group of 5 to 10

participants from the target audience is selected to play the game and provide feedback through observation and questionnaires. Their insights were used to refine gameplay, address bugs, and enhance the game's educational value.

By applying these methods and enhancements, the study aims to create a historically accurate and effective educational game that resonates with modern learners and contributes to cultural preservation through interactive media.

### Components and Design

The proposed game architecture for the click-and-drag game, developed in Godot, follows a structured, modular design that ensures efficient interaction between the user and the system components.

**Player Input.** In the initial stage, it serves as the primary trigger of the system. The player performs actions such as clicking and dragging using an input device (e.g., mouse or touchscreen). These actions are captured and forwarded to the system for processing.

**User Interface (UI).** It functions as the intermediary between the player and the game logic. The UI interprets the user's actions and determines the appropriate

response, such as selecting an object, dragging it across the screen, or releasing it at a specific location.

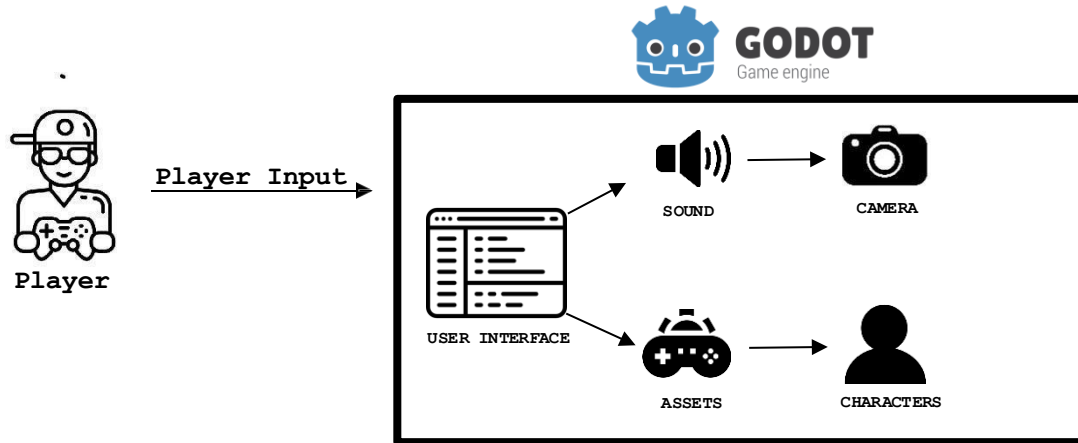
Assets. Subsequently, the processed input is directed to the asset management component, where game assets (e.g., sprites and interactive objects) are manipulated accordingly. These assets respond dynamically to user interactions, enabling movement, positioning, and interaction within the game environment.

The updated state of the assets is then reflected in the characters or game objects, ensuring that all visual and interactive elements accurately represent the player's actions in real time.

Sound. The architecture incorporates a sound system, which provides auditory feedback corresponding to user interactions, thereby enhancing the overall user experience. Simultaneously, the camera system is responsible for rendering and displaying the updated game scene, allowing the player to observe the results of their actions seamlessly.

Overall, the architecture shown in Figure 1 follows a linear yet interconnected flow. Player input is processed by the user interface, which updates the assets and game objects, while auxiliary systems such as sound and camera provide feedback and visualization. This design promotes

clarity, scalability, and maintainability in developing the click-and-drag game.



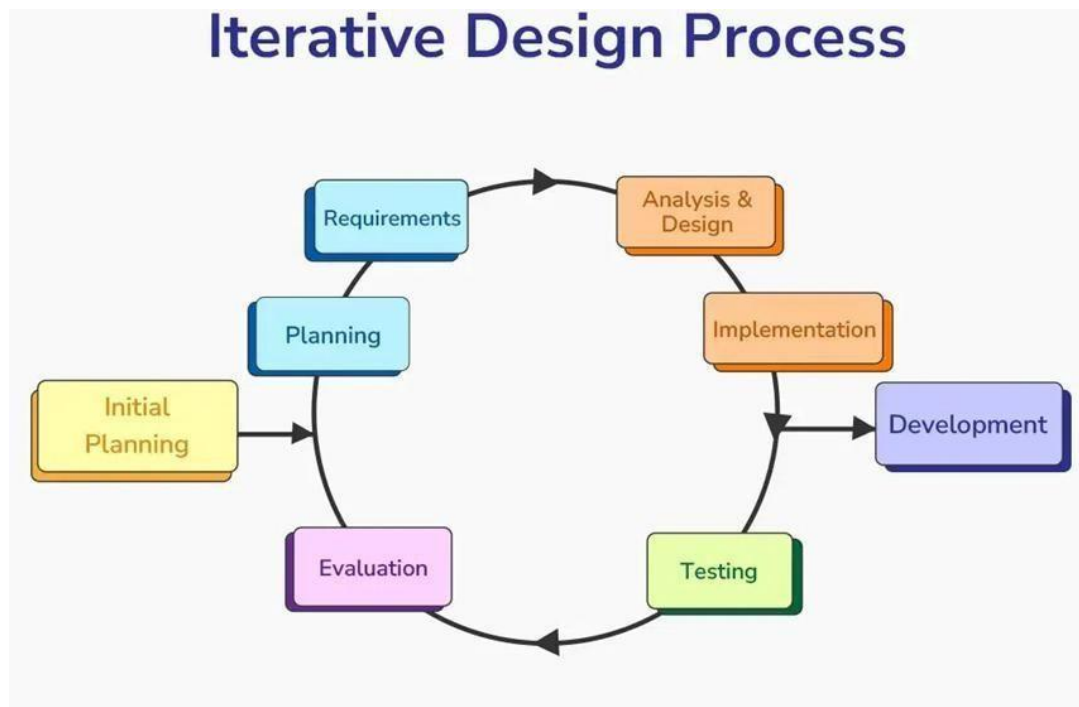
**Figure 1.** Game Architecture of the Proposed Game

The process follows a cyclical model consisting of planning and research, game design, development, and testing and evaluation as shown in Figure 2. In the initial phase, relevant historical data and core gameplay concepts are identified to establish the foundation of the game. This is followed by the game design phase, where mechanics, level structure, user interface, and asset requirements are defined.

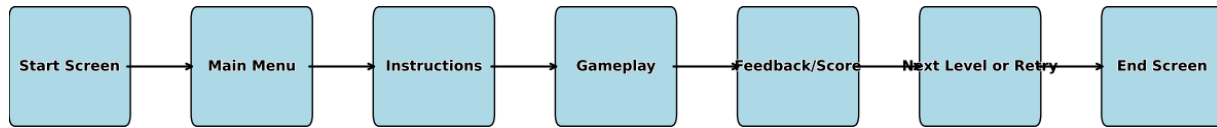
Subsequently, the development phase involves implementing these design elements in the selected game engine, integrating gameplay systems, visual assets, and audio components into a functional prototype. The

developed prototype then undergoes testing and evaluation, including alpha and beta testing, to assess functionality, usability, and overall player experience.

The results obtained from testing are used to inform necessary refinements and improvements. These modifications are reintroduced into the design and development phases, forming a continuous feedback loop. This iterative approach ensures that the game is progressively enhanced in terms of technical performance, usability, and educational effectiveness.



**Figure 2.** Game Development Cycle: Iterative Model



**Figure 3.** Game Flow of the proposed game

Start Screen. The game begins with the start screen, which serves as an initial visual introduction. It sets the thematic tone and allows the player to proceed to the main menu.

Main Menu. The main menu functions as the central hub of the game. Players are given options to start the game, access instructions, adjust settings, or exit the application. The interface is designed to be intuitive, ensuring ease of navigation.

Instructions. Before entering gameplay, players may access an instruction screen that explains the game mechanics, objectives, and controls. This phase ensures that users understand how to interact with the game and complete tasks effectively.

Gameplay. This is the core phase where the player engages with the game environment. Using click-and-drag mechanics, players manipulate characters and objects to achieve objectives aligned with the historical events of the Philippine Revolution. The system continuously

processes inputs, updates the game state, and provides visual and auditory feedback to support engagement and learning.

Feedback/Score. Upon completing a level, the player receives feedback on performance, including scores, progress indicators, and narrative outcomes. This phase reinforces learning, highlights accomplishments, and identifies areas for improvement.

Next Level or Retry. Based on performance, the player may proceed to the next level or retry the current stage. This phase maintains a sense of progression while providing opportunities to master game mechanics and content.

End Screen. After completing all levels or exiting, the game concludes with an end screen summarizing overall performance, achievements, and historical insights. This final phase provides closure and reinforces the educational objectives of the game.

Overall, this structured flow ensures that players experience a clear progression, combining engagement, educational content, and user-friendly navigation throughout the game.

## CHAPTER 4 RESULTS AND DISCUSSION

### Implementation

This section presents the implementation phases of the study, highlighting the pre-production, production, and post-production phases involved in developing the 2D click-and-drag game "*KKK: A Historically Based Click-and-Drag Game About the Philippine Revolution*" using the Godot Game Engine.

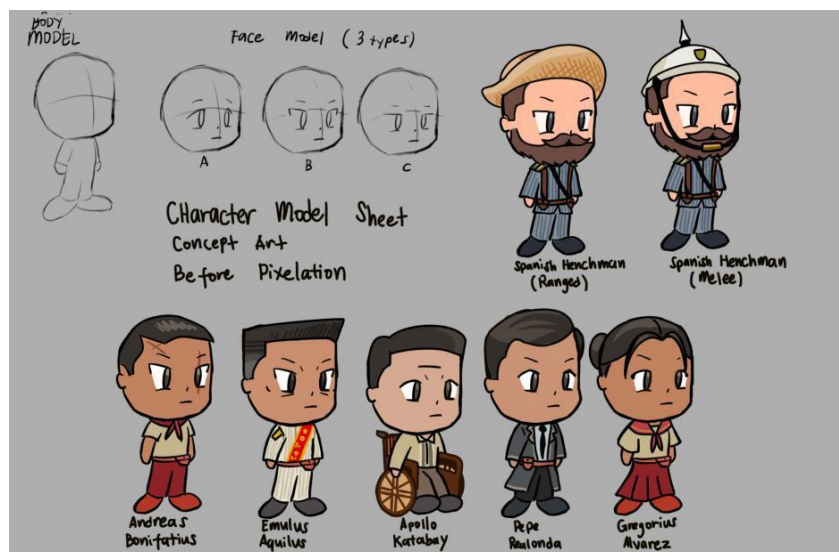
The pre-production phase focused on conceptual planning, including developing the game's historical narrative and the initial design of its mechanics. Game assets were designed using tools such as Aseprite for pixel art creation and Figma for user interface planning.

The production phase involved the creation and integration of 2D game assets, interactive mechanics such as drag-and-drop functionality, sound design, background music, and user interface components.

The concept art phase focuses on defining the game's visual identity by exploring proportions, poses, and stylistic features appropriate to a chibi aesthetic. This approach emphasizes simplified forms, exaggerated features, and expressive designs to enhance visual clarity and player engagement within a 2D environment.

Furthermore, the designs are created with consideration for their conversion into pixel art, ensuring that details, silhouettes, and proportions remain clear and recognizable when translated into a lower-resolution, grid-based format.

Figure 4 shows the rough illustrations that served as a guide for the subsequent asset production process, particularly for integrating the designs into pixel-based sprites. The concept art is refined and adapted into pixel art using appropriate tools, maintaining stylistic consistency while optimizing the assets for in-game implementation and animation. By establishing a clear visual direction early in development, the concept art supports coherence across all graphical elements and contributes to a cohesive user experience.



**Figure 4.** Assets Concept Art

Following the asset design stage, the rough sketches that are shown in Figure 5 were refined and converted into pixel-based sprites, ensuring consistency with the intended chibi aesthetic and compatibility with the game's 2D format. These finalized assets were then integrated into the system using the Godot Engine, where they were organized into scenes and nodes to enable interaction and functionality within the game.

In addition to character and environment assets, the user interface (UI) components—such as menus, buttons, and in-game overlays—were implemented to support navigation and gameplay interaction. The UI design emphasizes clarity, accessibility, and responsiveness, allowing players to interpret game information and perform actions easily.

This phase is closely tied to the overall design and development process. By incorporating the assets into the Godot Engine, the researchers were able to test their behavior, alignment, and responsiveness in real time, ensuring that both visual and functional aspects of the game operate cohesively. The implementation of assets and UI elements thus plays a critical role in transforming the design concepts into an interactive, playable system.



**Figure 5.** Assets Implementation and UI Design

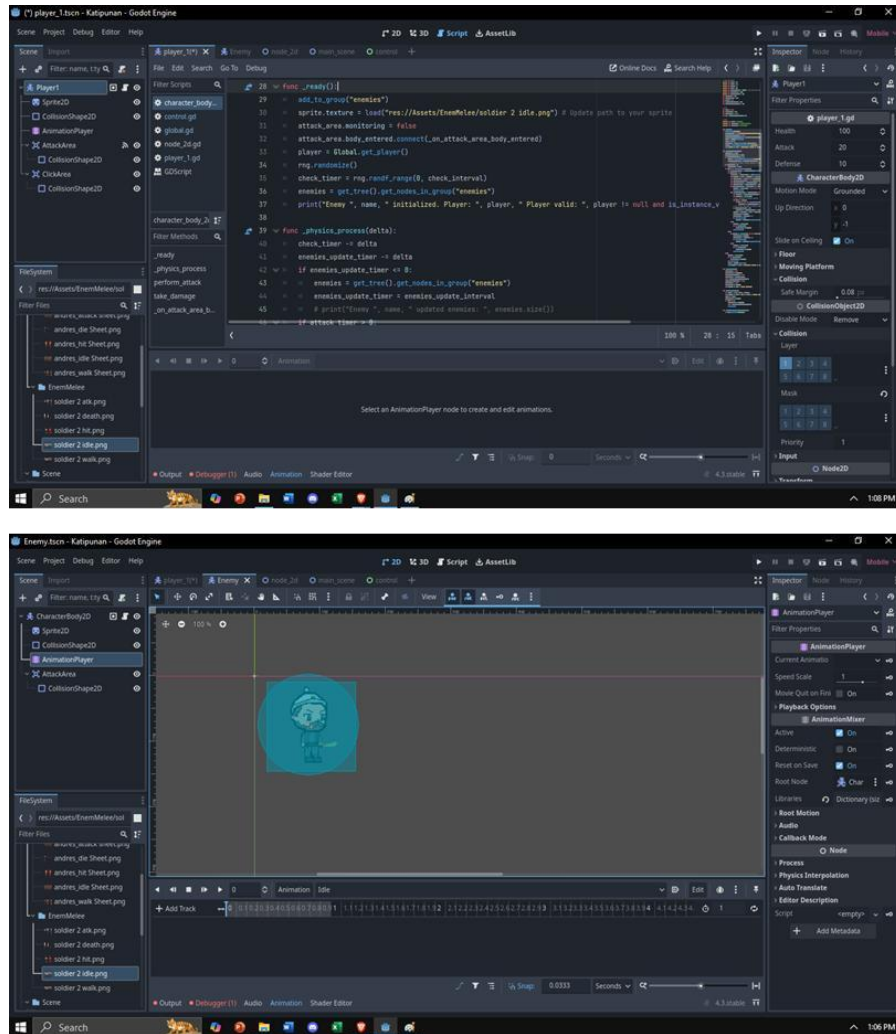
The integration of assets is performed incrementally, wherein individual sprites, animations, and interactive objects are assigned to specific nodes within the Godot scene system. Each asset is linked to a corresponding script that defines its behavior, such as movement, interaction, and response to player input. Through GDScript, properties such as position, visibility, collision, and state changes are controlled, allowing assets to function cohesively within the game environment.

User interface elements are also implemented through scripting, enabling dynamic interaction between the player and the system. Buttons, menus, and on-screen indicators

are connected to functions via signal-based events that trigger specific actions, such as starting the game, navigating menus, or updating on-screen information. This ensures that the UI remains responsive and synchronized with the overall game state.

For gameplay implementation, in Figure 6, the GDScript was used to handle core mechanics such as click-and-drag interactions, object manipulation, level progression, and feedback systems. Scripts continuously process player inputs, update the game state, and manage transitions between scenes, ensuring smooth, consistent gameplay flow.

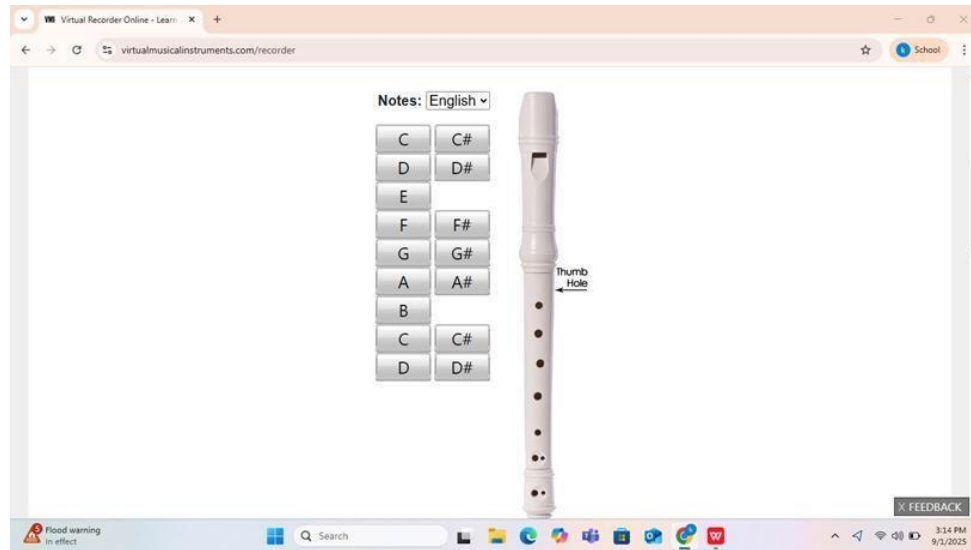
Overall, the use of GDScript within the Godot Engine enabled the systematic integration of assets and interface components, transforming static visual elements into interactive systems. This scripting process was essential for realizing the designed mechanics and ensured that all components of the game operate in a unified, functional manner.



**Figure 6.** GDScript Code

This phase involved composing and editing background music and sound effects using FL Studio, a digital audio workstation utilized for producing and refining the game's audio elements. The music production that is shown in figure 7 was designed to align with the game's theme and setting, supporting immersion and reinforcing the emotional tone of gameplay.

Once finalized, the audio assets are exported and integrated into the game using the Godot Engine, where they are synchronized with in-game events such as player interactions, scene transitions, and gameplay sequences. This ensures that the sound design complements the visual components and contributes to a cohesive and engaging user experience.



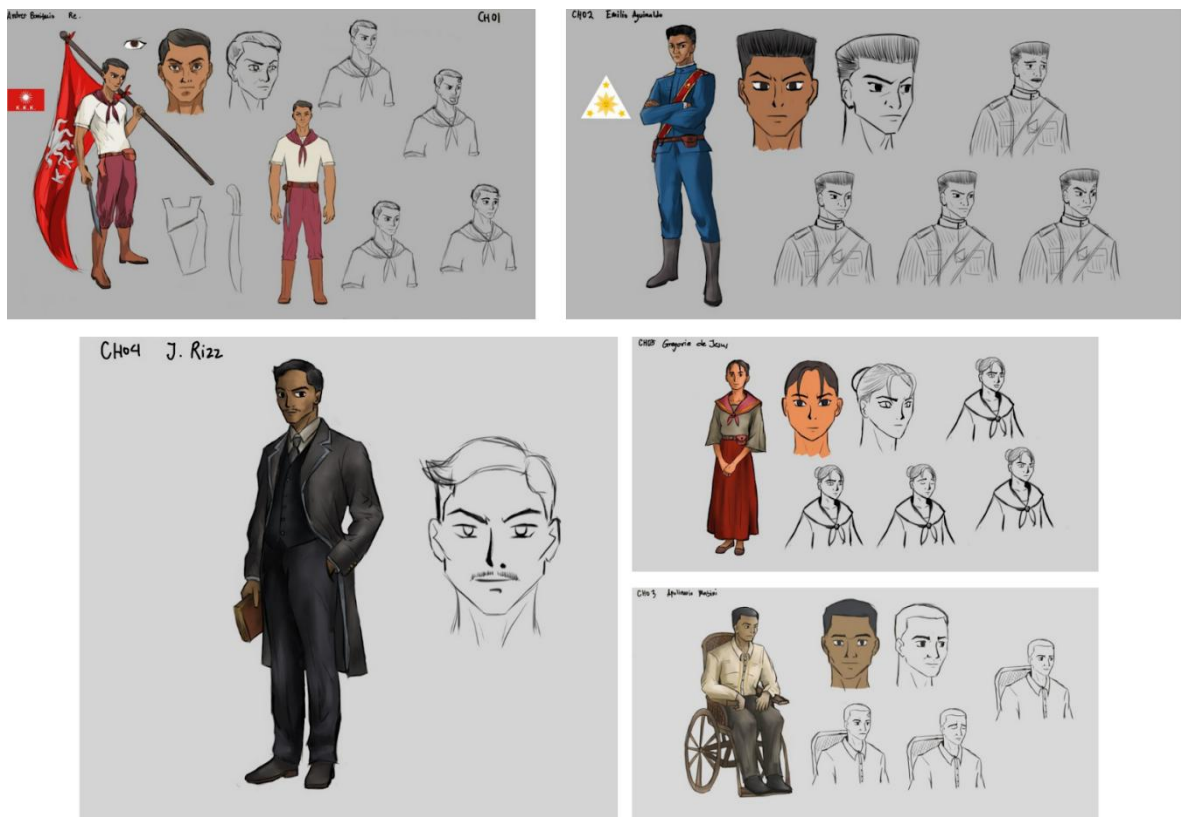
**Figure 7.** Music Production

The drawings shown in figure 8 were created using Autodesk SketchBook, allowing for detailed and expressive visual representations that align with the historical context of the Philippine Revolution.

The illustrations emphasize recognizable features, attire, and overall likeness of each figure while

maintaining a stylized aesthetic consistent with the game's visual direction. Unlike the in-game sprites, which follow a chibi and pixel-based style, these assets are intended specifically for cutscenes to enhance storytelling and narrative delivery.

These character illustrations are integrated into the game during cutscenes, supporting dialogue and key story moments. Their inclusion strengthens the narrative experience and reinforces the game's educational aspect by visually representing important historical figures.



**Figure 8.** Character Concept Art

The main menu provides access to essential functions such as starting the game, viewing instructions, adjusting settings, and exiting the application. It is designed to be visually clear and user-friendly, allowing players to understand and navigate the available options easily.

The layout incorporates thematic visual elements consistent with the game's overall design, ensuring coherence between the interface and the historical setting.

As shown in Figure 9, the interactive components such as buttons, are implemented in the Godot Engine and programmed in GDScript to respond to user input, enabling smooth transitions between game scenes.



**Figure 9.** Main Menu

The cutscene on Figure 10 uses the previously developed character concept art, displayed alongside dialogue to guide the player through key story moments. These sequences are designed to provide context, enhance immersion, and support the game's educational objectives by presenting historical information in a more engaging and accessible format.



**Figure 10.** In-Game Cutscene

The level selection shown in Figure 11 was divided into three primary sections, each corresponding to a major historical event: the Battle of San Juan del Monte, the Battle of Imus, and the Battle of Zapote Bridge.

Each battle was visually represented through a distinct background environment that reflects its setting,

such as landscapes, urban areas, or structural elements like bridges. These visual cues enhance immersion and help differentiate each level contextually. Within each battle, the interface provides three selectable stages labeled as Level 1, Level 2, and Level 3, representing progressive phases of gameplay. This staged progression allows players to experience increasing levels of difficulty and complexity within each historical scenario.

Interactive buttons are prominently displayed within each section, allowing players to select a specific level. Additional interface elements, such as "Back" buttons, are included to facilitate navigation and allow users to return to previous menus. Labels such as "Act" indicators further organize the progression structure, guiding players through the sequence of levels.



**Figure 11.** Level Selection

During gameplay, players utilize click-and-drag mechanics to control characters and interact with enemies to accomplish level objectives. Players assume the role of a notable hero from the Philippine Revolution. As they progress through the stages, both the number of characters and the number of enemies increase, indicating a gradual rise in difficulty over time. At the end of each stage, a battle takes place against a stronger enemy who represents the chief of the Spanish invaders. Upon defeating enemies, the main character and their allies may gain buffs that boost their morale points.

A key feature of the system is the morale mechanism, which reflects the overall state of the player's forces or public support. Player actions directly influence this

system, particularly through protecting civilians from the Spanish invaders. Successful decisions and interactions increase morale, while ineffective actions or losses may decrease it.

As respectfully shown on Figure 12, each character plays a distinct role in battle. A melee fighter typically wields a handheld weapon to attack enemies, while a ranged fighter uses a ranged weapon directly.

When engaging enemies in battle, surrounding a damaged hero with a healer can help gradually restore their health, enabling them to regain strength and continue fighting.

The interface provides real-time feedback through visual indicators such as health bars, morale levels, animations, and other on-screen elements, allowing players to monitor both their progress and morale throughout the game. Players may quit a level or revisit previously completed levels; however, progression is only possible once a level has been fully completed.



**Figure 12.** In-Game Cutscene

### Results Interpretation and Analysis

The evaluation employed a structured questionnaire consisting of Likert-scale items and open-ended questions. The Likert-scale items measured participants' perceptions of usability, performance, visual and audio quality, and overall gameplay experience on a 5-point scale (from Strongly Disagree to Agree Strongly). Open-ended questions were included to capture qualitative feedback regarding bugs, glitches, confusing sections, and suggestions for improvement.

Usability and Core Mechanics. Most respondents agreed that the core mechanics were clearly defined and that the controls responded reliably. However, several issues were noted with collision detection and click hitboxes, which sometimes made character targeting difficult. This suggests that while the drag-and-drop and point-and-click systems were functional, refinements in responsiveness and clarity are needed to improve gameplay readability.

Visual and Audio Assets. Feedback indicated that sprites and cutscenes were inconsistent in size, and the art style shifted between cartoon and pixel aesthetics, reducing immersion. On the audio side, sound effects were praised for clarity, but background music was either

missing or abruptly cut off, which weakened the atmosphere. These findings highlight the importance of maintaining visual consistency and balanced audio integration to support player engagement.

Level Design and Progression. Respondents criticized the repetitive level design, noting that levels felt similar except for increasing enemy counts. Some players also discovered they could skip levels, undermining the intended progression system. This indicates a weakness in game flow and progression mechanics, which should be addressed to sustain motivation and ensure meaningful learning sequences.

Technical Stability and Bugs. While the final build was generally stable, QA testing revealed bugs such as merging soldiers, unresponsive controls, and health bar visibility issues. These technical flaws disrupted gameplay and reduced usability. The presence of potentially copyrighted music assets were also flagged, raising concerns about compliance and long-term deployment risks.

Educational and Engagement Impact. Despite technical shortcomings, respondents acknowledged that the game's historically grounded decision-making mechanics—particularly the morale system, which required players to

manage public sentiment based on real historical events—actively reinforced contextual understanding of the period. These mechanics encouraged learners to consider cause- and-effect relationships, thereby supporting historical reasoning and retention. However, repetitive combat sequences and limited variation in mission objectives served primarily as engagement tools rather than instructional mechanisms, reducing sustained immersion and weakening long-term knowledge retention.

To provide context for the study, respondents were categorized into two main groups: students and non-students. Table 1 presents the proportion of each group within the sample of five individuals. As shown, students comprised the majority at 60%, while non-students accounted for the remaining 40%. This breakdown offers a clear view of the participants' demographic composition and serves as a foundation for interpreting subsequent findings

**Table 1.** Responder Type Distribution (5 respondents)

| RESPONDENT TYPE | PERCENTAGE |
|-----------------|------------|
| STUDENTS        | 60%        |
| NON- STUDENTS   | 40%        |

Table 1 shows the variety of respondents who volunteered for testing and evaluation. Of the 5 respondents, over 60% were students, and the remaining 40% non-students.

To guide the interpretation of survey responses, the study employed a 5-point Likert scale. Table 2 outlines the scoring range and corresponding qualitative categories, from "Strongly Disagree" to "Strongly Agree." This framework ensures that numerical averages can be consistently translated into meaningful interpretations, providing clarity in analyzing participant feedback and aligning quantitative scores with qualitative judgments.

**Table 2.** 5-Point Likert Scale scoring range

| INTERPRETATION    | VALUE | RANGE       |
|-------------------|-------|-------------|
| Strongly Agree    | 5     | 4.21 - 5.00 |
| Agree             | 4     | 3.41 - 4.20 |
| Neutral           | 3     | 2.61 - 3.40 |
| Disagree          | 2     | 1.81 - 2.60 |
| Strongly Disagree | 1     | 1.00 - 1.80 |

The primary instrument used in this study was a structured questionnaire designed to gather both demographic information and evaluative feedback on the game prototype. The questionnaire items were carefully constructed to assess key aspects of the game, including core mechanics, level design, controls, performance, and visual assets. Results of these evaluations are summarized in Tables 3 and 4, where mean scores and interpretations reflect participant perceptions of the game's strengths and areas for improvement.

**Table 3.** Evaluation of Results

| Component/Item                                             | Mean Score | Interpretation    |
|------------------------------------------------------------|------------|-------------------|
| Core mechanics were clearly defined                        | 2.00       | Disagree          |
| Level design followed a logical structure                  | 3.60       | Agree             |
| Controls responded accurately and reliably                 | 1.40       | Strongly Disagree |
| Game runs smoothly without major issues                    | 3.40       | Neutral           |
| Collision detection and physics implemented effectively    | 2.80       | Neutral           |
| UI behaves as intended                                     | 3.60       | Agree             |
| Asset integration (sprites, audio, UI) was smooth          | 3.40       | Neutral           |
| Visual Assets were produced consistently with the artstyle | 3.20       | Neutral           |
| Character animations enhanced gameplay readability         | 3.20       | Neutral           |

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|                                                       |      |         |
|-------------------------------------------------------|------|---------|
| Background music<br>fit the tone<br>and pacing        | 3.20 | Neutral |
| Sound effects<br>provided clear<br>feedback           | 3.60 | Agree   |
| Overall<br>technical<br>quality of the<br>final build | 3.40 | Neutral |

As shown in the evaluation tables, the overall results reflect a mix of agreement and neutrality across different aspects of the game prototype. Respondents strongly disagreed with the reliability of controls and disagreed with the clarity of core mechanics. At the same time, they agreed that level design was logical, the UI behaved as intended, and sound effects provided clear feedback.

Technical areas such as performance, collision detection, asset integration, visual assets, character animations, background music, and final build stability mostly fell into the neutral range, indicating divided opinions where some agreed but others disagreed or remained undecided. Taken together, these findings suggest that while the prototype demonstrated strengths in design elements like level structure, UI, and audio feedback, several core and technical systems were perceived as incomplete or inconsistent, which aligns with the fact

that certain mechanics were unfinished at the time of testing. Overall, the game was moderately well-received, with clear areas of strength but also opportunities for refinement.

Objective 1 was to design a gameplay mechanic inspired by the Philippine Revolution, focusing on morale (morale system). Respondents disagreed that the core mechanics were clearly defined ( $M=2.00$ ) and strongly disagreed with control reliability ( $M=1.40$ ). This reflects the unfinished state of the morale system during testing. Since the morale mechanic was intended as the central feature, its incomplete implementation led to unclear gameplay. While ratings were low, they highlight the need to fully develop and integrate the morale system to achieve the intended revolutionary theme and strengthen the game's identity.

Objective 2 was to create and integrate two-dimensional (2D) pixel art assets reflecting the colonial Spanish era. Respondents gave neutral ratings for visual asset consistency ( $M=3.20$ ) and character animations ( $M=3.20$ ), and rated asset integration neutral as well ( $M=3.40$ ). While some agreed the art style was consistent, others were undecided or disagreed, showing mixed perceptions. This suggests that the 2D pixel art assets

captured elements of the colonial Spanish era but lacked full cohesion and polish. The outcome highlights the need for refinement in asset production and integration to strengthen stylistic consistency and enhance immersion.

Objective 3 was to develop three playable game levels showcasing battles of the revolution. Respondents agreed that level design followed a logical structure ( $M=3.60$ , Agree), indicating that the three playable levels were coherent and well-organized. This positive reception shows that despite unfinished mechanics; the levels successfully conveyed the intended battles of the revolution. The outcome highlights level design as one of the prototype's strengths, effectively supporting the historical theme and gameplay flow.

Objective 4 was to evaluate the technical aspects of the game prototype. Respondents gave neutral ratings for overall performance ( $M=3.40$ ), collision detection and physics ( $M=2.80$ ), asset integration ( $M=3.40$ ), and final build stability ( $M=3.40$ ). These outcomes show that while the prototype was functional, technical systems lacked polish and consistency. The neutral stance highlights the need for optimization and refinement to ensure smoother gameplay and a more stable build.

## CHAPTER 5 SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

### Summary of the Proposed Study

This study proposes the development and evaluation of a 2D educational game entitled "*KKK: A Historically Click-and- Drag Game about the Philippine Revolution*". The project aims to integrate game development and gamification principles into history education, specifically focusing on the Philippine Revolution, to enhance student engagement, retention, and motivation.

The game is designed using the Godot Engine and employs intuitive click-and-drag mechanics to allow learners to interact with historical events, figures, and battles in a structured progression. Each level represents a milestone in the Revolution.

The study follows a development-first approach, emphasizing alpha and beta testing phases to identify bugs, refine mechanics, and validate usability before assessing educational effectiveness. Evaluation is conducted through a mixed-methods questionnaire that combines Likert-scale items and open-ended questions, capturing both quantitative measures of usability and qualitative feedback on the gameplay experience.

By situating the project within existing literature on gamification, usability testing, and educational game design, the study addresses gaps in localized, historically accurate, and technically robust learning tools. The expected outcome is a stable, engaging, and pedagogically sound game that demonstrates how gamified digital media can supplement traditional instruction in Philippine history.

### Summary of Findings

The findings confirm that the game prototype achieved basic usability and engagement goals, validating the feasibility of using Godot Engine for educational game development. At the same time, the results highlight the need for refinements in collision detection, level progression, and asset consistency to maximize both technical performance and educational effectiveness.

**Core Mechanics and Usability.** Respondents generally agreed that the game's core mechanics were clearly defined and that the controls responded reliably. However, they also noted issues with collision detection, targeting hitboxes, and character responsiveness, which at times disrupted the overall gameplay experience.

**Visual and Audio Assets.** The integration of sprites, models, and animations was functional, but several participants observed inconsistencies in the art style, such as the mix of cartoon and pixel aesthetics and variations in cutscene sprite sizes. While sound effects were praised for their clarity and effective feedback, the background music was either missing or abruptly cut off, which reduced overall immersion.

**Level Design and Progression.** Levels followed a logical structure but were perceived as repetitive,

offering little variation beyond increasing enemy counts. Additionally, some respondents discovered that levels could be skipped, which undermined the intended progression system and disrupted the education flow.

Technical Stability and Bugs. The final build was generally stable across devices, but bugs were identified during QA testing, including merging soldiers, unresponsive controls, and health bar visibility issues. Concerns were raised about the use of potentially copyrighted music assets, which could pose risks for deployment.

Educational and Engagement Impact. Respondents found the game's premise unique and engaging despite technical shortcomings. The morale system and historically grounded decision-making mechanics supported learning by reinforcing cause-and-effect relationships in historical events. These mechanics contributed to contextual understanding and improved retention of historical concepts.

In contrast, repetitive combat sequences and limited mission variety primarily served as engagement tools rather than instructional elements. These repetitive mechanics reduced sustained immersion and limited deeper

cognitive processing. Overall, mechanics aligned with learning objectives were more effective in promoting retention than those designed solely for entertainment.

The Participants may have evaluated the Overall Technical Quality more holistically, considering aspects such as graphics, performance consistency, and general playability, while still noting specific technical flaws in their qualitative feedback. In other words, the high mean score reflects a positive overall impression despite notable but isolated issues.

## Conclusions

This study set out to design, develop, and evaluate a 2D educational game that integrates gamification principles into the teaching of Philippine history, specifically the Revolution. Using the Godot Engine, the game employed click-and-drag mechanics to allow learners to interact with historical events and figures in a structured, engaging way, providing a platform for history learning.

Respondents agreed that the game's core mechanics were intuitive and that its interactive format encouraged participation. These findings align with existing literature on gamification, which emphasizes the motivational benefits of immediate feedback and interactive design.

At the same time, the study revealed areas requiring refinement. Issues in collision detection, inconsistent art styles, repetitive level design, and minor bugs limited the overall usability and immersion. Concerns about asset consistency further highlight the need for careful production workflows in educational game development.

Despite these limitations, the project confirms the potential of gamified digital tools to supplement traditional instruction in Philippine history. By combining usability testing with pedagogical design, the

study contributes to the growing body of research on educational game development in local contexts. Future iterations should focus on improving technical polish, diversifying gameplay, and strengthening narrative integration to maximize both engagement and learning outcomes.

### Recommendations

Technical Refinement. Improve collision detection and targeting hitboxes to ensure smoother character control and responsiveness, and address bugs identified during QA testing, such as merging soldiers, health bar visibility issues, and unresponsive controls. Ensure all audio and visual assets are consistent in style and properly optimized to avoid performance issues.

Game Design Enhancements. Diversify level design to reduce repetition and sustain player engagement. Incorporating varied objectives, environments, or challenges can strengthen immersion and the progression system by preventing level skipping and ensuring that each stage builds meaningfully on the previous one. Additionally, integrating narrative depth through contextual prompts, cutscenes, or historical storytelling can reinforce learning outcomes.

Educational Integration. Align gameplay mechanics more closely with learning objectives, ensuring that historical facts and events are reinforced through interaction, and include in-game hints or tutorials to guide players who struggle with mechanics, and improving accessibility for younger learners. Additionally, develop assessment features to measure retention of historical knowledge after

gameplay.

Production and Compliance. Establish a consistent art and audio pipeline to maintain coherence across all assets and replace or re-license any potentially copyrighted music or assets to avoid legal risks in future deployment. Additionally, document workflows and asset sources thoroughly to support transparency and reproducibility.

Future Research Directions. Conduct expanded usability testing with larger and more diverse participant groups to validate findings and explore the integration of adaptive difficulty systems to personalize learning experiences. Additionally, investigate the long-term impact of gamification on students' retention of historical knowledge by comparing game-based learning with traditional methods.

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## Appendices

## Appendix A. Source Code

```
35 func _input(event: InputEvent) -> void:
36     if not is_selected or is_dead: return
37
38     if event.is_action_pressed("click"):
39         var mouse_pos = get_global_mouse_position()
40         var hit = _get_clicked_enemy(mouse_pos)
41         if hit:
42             target_enemy = hit
43             print("Targeted: ", target_enemy.name)
44         else:
45             move_target = mouse_pos
```

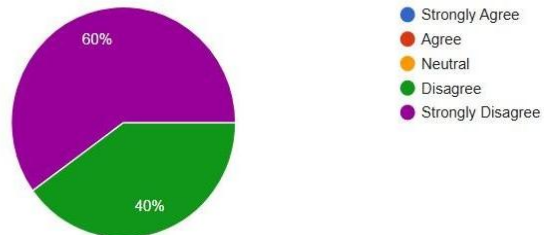
```
125
126 func _get_clicked_enemy(pos: Vector2) -> Node2D:
127     var query = PhysicsPointQueryParameters2D.new()
128     query.position = pos
129     query.collision_mask = 2
130     var hits = get_world_2d().direct_space_state.intersect_point(query)
131     for hit in hits:
132         if hit.collider.is_in_group("enemies"):
133             return hit.collider
134     return null
135
```

```
1 extends CharacterBody2D
2
3 @export var max_health: int = 150
4 @export var speed: float = 200.0
5 @export var attack_range: float = 150.0
6 @export var attack_damage: int = 25
7 @export var attack_cooldown: float = 1.0
8
9 var health: int = max_health
10 var is_dead: bool = false
11 var is_selected: bool = false
12 var move_target: Vector2
13 var target_enemy: Node2D = null
14
15 @onready var sfx_hit: AudioStreamPlayer2D = $sfx_hit
16 @onready var sfx_death: AudioStreamPlayer2D = $sfx_death
17 @onready var sprite: AnimatedSprite2D = $Sprite2D
18 @onready var taunt_timer: Timer = Timer.new()
19 @onready var attack_timer: Timer = Timer.new()
```

## Appendix B. Raw Data and Participant Feedbacks

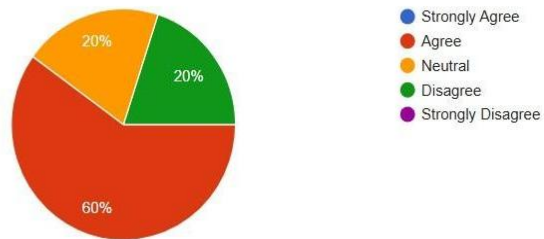
3. The game's controls respond accurately and reliably.

5 responses



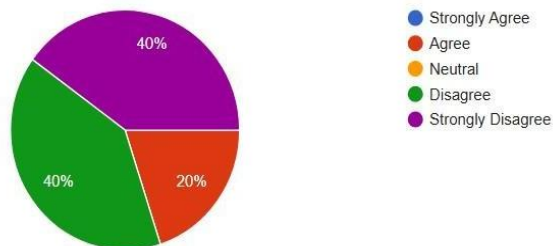
4. The game runs smoothly without major performance issues.

5 responses



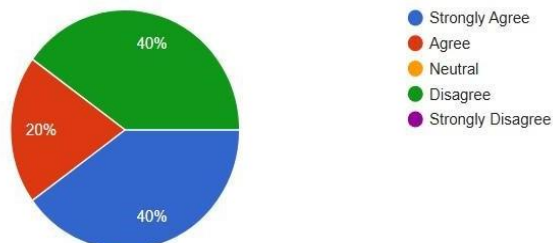
1. The game's core mechanics were clearly defined.

5 responses



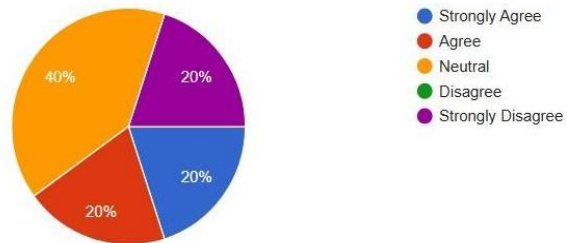
2. Level design followed a logical and consistent structure.

5 responses



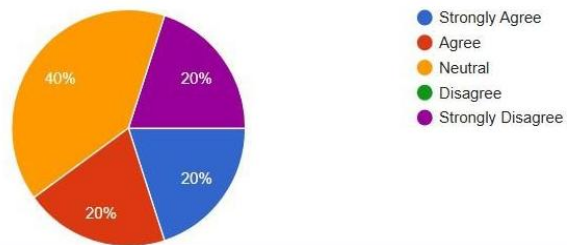
8. The visual assets (textures, sprites, models) were produced consistently with the art style.

5 responses



9. Character animations run smoothly and enhance gameplay readability.

5 responses



### Appendix C. Evaluation Questionnaire

1. The game's core mechanics were clearly defined. \*

*Mark only one oval.*

- ☐ Strongly Agree  
☐ Agree  
☐ Neutral  
☐ Disagree  
☐ Strongly Disagree

3. The game's controls respond accurately and reliably. \*

*Mark only one oval.*

- ☐ Strongly Agree  
☐ Agree  
☐ Neutral  
☐ Disagree  
☐ Strongly Disagree

2. Level design followed a logical and consistent structure. \*

*Mark only one oval.*

- ☐ Strongly Agree  
☐ Agree  
☐ Neutral  
☐ Disagree  
☐ Strongly Disagree

4. The game runs smoothly without major performance issues. \*

*Mark only one oval.*

- ☐ Strongly Agree  
☐ Agree  
☐ Neutral  
☐ Disagree  
☐ Strongly Disagree

## **Appendix D. Game Narrative**

Long ago, life was better for the Filipinos, which before the Spanish era, were known as Indios, they lived on an island, in peace and harmony. They took care of the land, and the land gave them everything they needed. In the eyes of the Indios, everything was simple and good.

Until one day, their generational peace was met with steel and fire, as they watch the invading Spaniards vying for resources, eyes filled with greed, ultimately subjugated the island and its people by the Empire of Spain, ultimately losing their identity.

Here, the story of the Andres Bonifacio starts. For years he has trained his countrymen in the arts of war, founded the Katipunan on Manila, and inspired its people, feeding their already insatiable fervor on wanting to gain independence from their cruel overlords, vowing to the People that they will rise and stand united against a Tyrannical superpower.

### **ACT 1: BATTLE OF SAN JUAN DEL MONTE**

Another usual gathering for Andres Bonifacio and other well-known names like Emilio Aguinaldo, Apolinario Mabini, Gregoria De Jesus y Alvarez, and many more. They're driving a plot on revolting against their Spanish Overlords, but is pointed out by Aguinaldo that they are woefully unprepared, as they have insufficient support from the people; resulting in low morale for most of the time, insufficient weapons to sustain the revolution, an unorganized command chain, and untrained leaders. If they were to have a successful revolution, they must overcome this insurmountable obstacle first.

Apparently, this was overheard by a whistleblower for the Spaniards, reporting it to the nearest Capitán, who then immediately dispatched a few Guardia Civil to investigate. As Bonifacio and the others were still bickering, they were immediately notified by a couple of farmers about an incoming Guardia Civil. Bonifacio and Aguinaldo looked at each other, as if trying to communicate to each other that this might be the only opportunity for them to successfully spark a revolution.

Bonifacio and the other Katipuneros that can wield a weapon, staged an ambush, each armed with bolos and a spark of fire in their eyes. A few hours passed when the Teniente and a few Guardia Civil arrived, and what they saw was none other than Bonifacio, each exchanged words with fiery temperament, with the latter doing so as to buy enough time for his Katipuneros to get into position.

Just as soon as the Teniente pulled out a Colt Model 1894, Bonifacio gave a signal by uttering a line "Kahit na kailangan kong mamatay para lang makatikim kami ng kahit kaunting kalayaan, kung gayunman", signaling the Katipuneros to attack the Guardia Civil, therefore the kickstarting the Revolution.

The aftermath of the ambush resulted in a few Katipunero Casualties, and the entirety of the Guardia Civil killed, with one soldier managing to get away and alerting the Gobernador Heneral, putting the entire island on high alert, with Cavite being the major center of revolutionary activity, experiencing significant battles between Filipino and Spanish Forces.

The news quickly spread to other provinces with Manila and Cavite the first to go, followed suit was Bulacan, Pampanga, Tarlac, Laguna, Batangas, and Nueva Ecija, this would become one of the key elements to the creation of the Philippine Flag.

After the discovery of the Katipunan, Andres Bonifacio became aware of the Spanish government's plans for military action. Bonifacio deployed several of his men around the Pasong Tamo bridge when he heard infantrymen and the Guardia Civil coming to raid communities around the bridge.

The revolutionaries regrouped at Santa Mesa and engaged the arriving Spanish troops. The regiment, together with the garrison, almost wiped-out Bonifacio's men, inflicting severe and heavy losses, something that Bonifacio could never recover, this disastrous outcome would force Bonifacio to retreat towards the Pasig River. It's estimated that the Katipuneros suffered around 150 killed, with 200 being captured, whilst the Spanish Army suffered only 2 casualties, with several wounded.

#### **END OF ACT 1.**

#### **ACT 2: BATTLE OF IMUS**

After years of peaceful life upended by Spanish conquest, Andres Bonifacio and the Katipunan ignite a revolution to reclaim Filipino freedom, the conflict erupting first at San Juan del Monte where a desperate ambush sparks widespread uprisings across Luzon.

Despite heavy losses and setbacks that force Bonifacio to retreat, the rebellion finds new strength in Cavite under Emilio Aguinaldo's leadership. There, carefully planned ambushes, use of local terrain, and coordinated strikes culminate in the Battle of Imus, where Katipunero forces seize the estate and bridge, shattering Spanish control in the region and inspiring recruits from neighboring provinces.

The victory at Imus, hard-won and costly, shifts momentum to the revolutionaries, forcing Spanish authorities to reinforce and reorganize while galvanizing broader resistance that transforms a scattered insurgency into a sustained struggle for independence.

#### **END OF ACT 2.**

### **ACT 3: BATTLE OF ZAPOTE BRIDGE**

From a time of simple harmony before Spanish conquest, Filipino communities lose their freedom and identity under colonial rule, sparking Andres Bonifacio to form the Katipunan and train compatriots for revolt.

The uprising erupts violently at San Juan del Monte, where costly ambushes and heavy losses force a tactical retreat but also ignite province-wide resistance, and leadership in Cavite under Emilio Aguinaldo refocuses the struggle into coordinated guerrilla tactics and decisive local actions. Imus becomes a pivotal victory when Katipuneros seize the estate and bridge through ambush, local knowledge, and daring charges, galvanizing recruits and undermining Spanish control.

The revolution's momentum carries into the decisive engagement at Zapote Bridge, where prepared defenses, obstacles, and close-quarters fighting repel Spanish assaults, inflict severe casualties on colonial forces, and consolidate revolutionary gains in Cavite, transforming scattered uprisings into a sustained, organized movement for independence despite heavy costs and lasting uncertainty about the road ahead.

**END OF ACT 3, NEXT STAGES COMING SOON**

## Appendix E. Character Dialogue Script

### Scene 1: The Gathering

(A dimly lit room. Bonifacio, Aguinaldo, Mabini, and Gregoria de Jesús sit around a table. The air is tense.)

**Aguinaldo:** “We are not ready, Andres. Our men lack weapons, training, even morale. If we march now, we march to our deaths.”

**Bonifacio (leaning forward):** “Emilio, if we wait for perfection, we will wait forever. The people suffer every day under Spanish rule. Better to fight with bolos than to die with empty hands.”

**Mabini (calmly):** “Courage without preparation is dangerous. But hesitation kills revolutions. We must decide now.”

**Gregoria de Jesús:** “The Guardia Civil will not wait for us to be ready. They will come. And when they do, we must show them we are not afraid.”

(Suddenly, farmers rush in, breathless.)

**Farmer:** “Kapitan! The Spaniards are coming — the Guardia Civil march this way!”

(Bonifacio and Aguinaldo exchange a silent, knowing look.)

### Scene 2: The Ambush

(Hours later. The Teniente and Guardia Civil arrive. Bonifacio steps forward, bolo in hand.)

**Teniente (mocking):** “Andres Bonifacio. You dare gather armed men against the Crown? Lay down your weapons, or face the Empire’s wrath.”

**Bonifacio (steady, fiery):** “Kahit na kailangan kong mamatay para lang makatikim kami ng kahit kaunting kalayaan... kung gayunman!” (He raises his bolo, signaling the Katipuneros.)

(The Katipuneros erupt from hiding, charging with bolos. The Guardia Civil fire back. Chaos ensues.)

### Scene 3: Aftermath

(Smoke clears. Katipuneros mourn their fallen. The Guardia Civil lie dead, save one soldier fleeing into the distance.)

**Aguinaldo (breathing heavily):** “We have won... but at a terrible cost. Too many of our brothers lie dead.”

**Bonifacio (grimly):** “Their blood waters the soil of freedom. Today, the revolution lives.”

**Gregoria de Jesús (resolute):** “Then let Cavite, Bulacan, Pampanga, and all of Luzon hear of this. Let the flame spread.”

#### **Scene 4: Santa Mesa Disaster**

(Days later. Bonifacio rallies his men at Santa Mesa. The sound of Spanish troops marching grows louder.)

**Bonifacio (to his men):** “Hold the line! For the Katipunan, for our freedom! Do not falter!”

(Gunfire erupts. The Katipuneros fall one by one. Mabini shouts over the chaos.)

**Mabini:** “We are being overrun! Retreat, Andres! Save who you can!”

**Bonifacio (anguished, shouting):** “No! We cannot retreat — not now!” *(But as his men are slaughtered, he realizes the truth. He grips his bolo tightly, then lowers it.)* “...Pasig River. Fall back! Live to fight another day!”

(The Katipuneros retreat, battered and broken. The Spanish stand victorious, their casualties few. Bonifacio watches, grief-stricken, as the revolution’s first great defeat unfolds.)

### **ACT 2:**

#### **Scene 1: Cavite Planning**

(Inside a Cavite estate. Aguinaldo, Bonifacio, and Katipuneros gather around a map spread across a wooden table.)

**Aguinaldo (pointing to the bridge):** “This is where we strike. The Spaniards will expect us to face them head-on, but we will use the riverbanks and the estate walls. Surprise is our weapon.”

**Bonifacio (nodding):** “Your plan is sound. Cavite knows its land better than any soldier from Madrid. Let us turn their arrogance into their downfall.”

**Katipunero Scout:** “The Guardia Civil march at dawn. They are confident, but they do not know we wait.”

#### **Scene 2: The Ambush**

(Spanish troops approach the Imus bridge. Aguinaldo signals his men hidden among the trees and estate walls.)

**Spanish Officer (smirking):** "Filipino rebels with bolos think they can defeat Spain? Crush them quickly."

**Aguinaldo (raising his hand, whispering):** "Hold... hold... now!"

(Gunfire and bolos erupt. Katipuneros swarm from cover, cutting off the bridge. The Spaniards falter as the terrain traps them.)

**Bonifacio (shouting over the chaos):** "Push them back! For our fallen brothers at San Juan del Monte!"

### **Scene 3: Turning the Tide**

(The Spaniards regroup, but Katipuneros seize the estate and cut off reinforcements. Aguinaldo confronts the Spanish officer.)

**Spanish Officer (furious):** "You peasants dare defy the Crown? You will be crushed!"

**Aguinaldo (firm, resolute):** "We are not peasants. We are Filipinos. And today, Cavite is free."

(The officer is overwhelmed as Katipuneros secure the bridge. The Spaniards retreat in disarray.)

### **Scene 4: Aftermath**

(Katipuneros cheer, though many lie wounded. Aguinaldo surveys the battlefield, solemn yet proud.)

**Bonifacio (placing a hand on Aguinaldo's shoulder):** "You have given the revolution new life. Imus will be remembered."

**Aguinaldo (quietly, to his men):** "This victory is not mine alone. It belongs to every Filipino who dares to dream of freedom. Today, we show the world that Spain can bleed."

**Katipuneros (chanting):** "Mabuhay ang Kalayaan! Mabuhay ang Katipunan!"

## **ACT 3**

### **Scene 1: Preparing the Defenses**

(Zapote Bridge. Aguinaldo and Bonifacio stand with Katipuneros, overseeing barricades of carts, logs, and sharpened bamboo.)

**Aguinaldo (pointing to the bridge):** “Here we make our stand. The Spaniards will come with rifles and cannons, but this bridge will be their grave.”

**Bonifacio (gripping his bolo):** “We have lost too many brothers at San Juan. Today, we fight not only to hold Cavite, but to prove we will never kneel again.”

**Katipunero Soldier:** “The barricades are ready, Kapitan. The riverbanks are lined with traps. They will not pass easily.”

### **Scene 2: The Spanish Assault**

(Spanish troops march forward, drums beating. A Spanish officer shouts orders as volleys of gunfire echo across the bridge.)

**Spanish Officer:** “Forward! Crush these rebels! Show them the might of Spain!”

(Gunfire rattles. Katipuneros crouch behind defenses, returning fire with rifles and charging with bolos when Spaniards breach the barricades.)

**Aguinaldo (commanding):** “Hold the line! Aim for the officers — break their command!”

**Bonifacio (to his men):** “Close the gap! Drive them back with steel and courage!”

(The fighting turns brutal, close-quarters on the bridge. Smoke and cries fill the air.)

### **Scene 3: Turning the Tide**

(Spanish forces falter as traps and barricades slow their advance. Katipuneros surge forward, cutting down soldiers in fierce combat.)

**Spanish Officer (panicked):** “Fall back! Fall back! The rebels are too strong!”

**Aguinaldo (raising his sword):** “Do not let them escape! Every step they retreat is a step toward our freedom!”

(The Spaniards retreat, leaving bodies scattered across the bridge. Katipuneros cheer, though many lie wounded.)

### **Scene 4: Aftermath**

(The battlefield is quiet. Aguinaldo and Bonifacio stand among their men, exhausted but victorious.)

**Bonifacio (solemnly):** “We have paid dearly. Too many names will be remembered only in silence.”

**Aguinaldo (resolute):** “But their sacrifice has given Cavite strength. Zapote Bridge is ours, and with it, hope for the nation.”

**Katipuneros (chanting together):**

“Mabuhay ang Kalayaan! Mabuhay ang Katipunan!”

## Appendix F. Music Score



**West Visayas State University**  
**COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY**  
**La Paz, Iloilo City, Philippines**

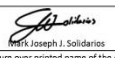
**Appendix G. Letter to the Advisor**

Attachment 3

|                                                                                   |                                                                                  |                     |                            |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------|----------------------------|
|  | <b>INVITATION LETTER FOR ADVISER</b><br><br><b>WEST VISAYAS STATE UNIVERSITY</b> | Document No.        | <b>WVSU-ICT-SOI-03-F03</b> |
|                                                                                   |                                                                                  | Issue No.           | <b>1</b>                   |
|                                                                                   |                                                                                  | Revision No.        | <b>0</b>                   |
|                                                                                   |                                                                                  | Date of Effectivity | <b>April 27, 2018</b>      |
|                                                                                   |                                                                                  | Issued by:          | <b>CICT</b>                |
|                                                                                   |                                                                                  | Page No.            | <b>Page 2 of 2</b>         |

PS:

*Advisors, are task to work with the students in providing direction and assistance as needed in their thesis/capstone project. They shall meet with the students weekly or as needed to provide direction, check on progress and assist in resolving problems until such a time that the students passed their defenses and submit their final requirements, as well as, preparing their evaluations and grades.*

|                                                                                                                          |                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Action Taken:</b><br><input checked="" type="checkbox"/> I Accept.<br><input type="checkbox"/> Sorry, I don't accept. | <br>Mark Joseph J. Solidarios<br>Signature over printed name of the Adviser |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

CC:

CICT Dean  
 Research Coordinator  
 Group  
\*To be accomplished in a space

Attachment 3

|                                                                                    |                                                                                  |                     |                            |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------|----------------------------|
|  | <b>INVITATION LETTER FOR ADVISER</b><br><br><b>WEST VISAYAS STATE UNIVERSITY</b> | Document No.        | <b>WVSU-ICT-SOI-03-F03</b> |
|                                                                                    |                                                                                  | Issue No.           | <b>1</b>                   |
|                                                                                    |                                                                                  | Revision No.        | <b>0</b>                   |
|                                                                                    |                                                                                  | Date of Effectivity | <b>April 27, 2018</b>      |
|                                                                                    |                                                                                  | Issued by:          | <b>CICT</b>                |
|                                                                                    |                                                                                  | Page No.            | <b>Page 1 of 2</b>         |

January 22, 2025

**MARK JOSEPH J. SOLIDARIOS**  
 Instructor

WEST VISAYAS STATE UNIVERSITY  
 Luna St, La Paz, Iloilo City, 5000 Iloilo

Dear Mr. Solidarios,

The undersigned are BS Entertainment and Multimedia Computing Research 1/Thesis 1 students of CICT, this university. Our thesis/capstone project title is *Kronika ng Kultura at Kasaysayan*.

Knowing of your expertise in research and on the subject matter, we would like to request you to be our **ADVISER**.

We are positively hoping for your acceptance. Kindly check the corresponding box and affix your signature in the space provided. Thank you very much.

Respectfully yours,

1.   
Paul Ernest Duraza
2.   
Kyle Francis Concepcion
3.   
Aaron Seth Joseph Purigay
4.   
Kent Lynne Aquista
5.   
Thomas Dominic Espinosa

**Appendix H. Disclaimer**

This research project and its corresponding documentation entitled "KKK: A HISTORICALLY BASED CLICK-AND-DRAG GAME ABOUT THE PHILIPPINE REVOLUTION" is submitted to the College of Information and Communications Technology, West Visayas State University, in partial fulfillment of the requirements for the degree, Bachelor of Science in Entertainment and Multimedia Computing. It is the product of our own work, except where indicated text.

We hereby grant the College of Information and Communications Technology permission to freely use, publish in local or international journal/conferences, reproduce, or distribute publicly the paper and electronic copies of this software project and its corresponding documentation in whole or in part, provided that we are acknowledged.

Kent Lynner P. Acuesta

Kyle Francis D. Concepcion

Paul Ernest B. Dureza

Thomas Dominic S. Espinosa

June 2026